VIRAJ CHOUDHARY

B.Tech Artificial Intelligence | 2023 - 2027
github.com/virajchoudhary

Education

Mukesh Patel School of Technology Management & Engineering, NMIMS 2023 - 2027

- B.Tech Artificial Intelligence
- Semester 6 CGPA: 3.25/4.00

Billabong High International School 2021 - 2023

- Class XII, CBSE: 67.60

Billabong High International School 2021

- Class X, CBSE: 81.80

Projects

AtlasOps - Multi-Agent SRE Research Platform | 2026

- Extended an inherited four-agent SRE platform with controls for proposed Kubernetes actions.
- Built authenticated, digest-bound FastAPI approval/history APIs and identity/audit-integrity tests.

Kerquant - Financial Analysis & Research Terminal | 2026

- Built a Next.js/TypeScript and Django terminal with private watchlists and saved research replay.
- Added isolated synthetic-demo startup; recorded 134 frontend checks and 17 Chromium scenarios passing.
- Contributed PyTorch LSTM/ARIMA/GARCH research and backtests; no production release.

GridShift - Grid Forecasting and GPU Scheduling Simulation | 2026

- Reported team SARIMA mean MAE 16.03% below persistence across five horizons on the frozen 2025 test.
- Reported 12.41% less modeled CO2 for 2,913 GPU jobs with team calendar-mean vs earliest scheduling; no deadline misses.

Internships

Kotak Mutual Fund | Software Engineering Intern | May 2026 - Present

- Developed Python/FastAPI and React dashboards for internal sales, financial analytics and MIS reporting.
- Built Excel/CSV validation and REST reporting exports; added regression tests and security controls.

Certifications & Publications

LEARN-Net - Explainable Diabetic Retinopathy Grading | 2025 - Present

Co-author | SN Computer Science manuscript under revision

- Co-authored four-class APTOS research using LAB-CLAHE, residual CNN/CBAM and Grad-CAM.
- Reported team-manuscript QWK 0.9097 and 82.51% accuracy with 2.26M parameters; no external clinical validation.
- Added FastAPI health/prediction/upload tests and React prediction/Grad-CAM API integration.

Double-Heston Bitcoin Option Pricing | 2026

Co-author | Current research manuscript

- Built a Fourier pricer and independent benchmark; 36/36 controlled cases passed in historical validation.
- Co-authored a fixed-parameter PINN study of 1,102 authentic Deribit trades in ten retrospective episodes; RMSE \$0.34-\$0.59 vs Fourier prices. No market calibration or trading signal.

AWS Academy Graduate: Cloud Architecting, April 2026; Cloud Foundations, March 2026

Core Skills

Technical Skills: Python, C++, JavaScript, TypeScript, SQL; PyTorch, scikit-learn, Pandas, NumPy, SciPy

Development: FastAPI, Django REST Framework, Next.js, React, Streamlit; computer vision, time-series modeling

Infrastructure: Git, GitHub Actions, Docker, Jenkins, Kubernetes, AWS, Linux

Non-Technical Skills: Cross-functional collaboration

Achievements and Positions of Responsibility

- Authored 12 merged codex-router PRs for quota failover, subagent lifecycle and tool-schema fixes.
- Submitted OpenCode #62753: fixed callable Standard Schema validation with regressions; not merged.